Computer Networks

ASSIGNMENT 1

CO1: *Apply the functions of the OSI layers in enabling reliable data transmission across networks – uses of networks, network hardware and software, components, types of connections, topologies, protocols and standards, the ISO/OSI reference model, transmission media, wireless communication, Ethernet configuration, an introduction to 5G networks, and error detection, error correction and data-link control.*

Assignment Title: Make the Data of a Smart Manufacturing Plant Arrive Correctly and On Time (Total Marks - 50)

Nature of Assignment: Individual
Duration: 1 Week
Submission Date: 02/11/2026

Bloom's Levels Assessed: K5 – Evaluate (every task asks you to calculate or analyse first, and then judge, decide or justify)

Mandatory Execution and GitHub Submission

- **Every student must individually execute every task in this assignment**, including all "Complete the Code" programs.
- Create a GitHub repository named **CSA07-<RegisterNumber>-Assignment1** and share it with the course faculty (public, or private with the faculty added as a collaborator).
- Keep one folder for each task (for example: Task01, Task02 … or Q1, Q2 …), containing the Python programs, Packet Tracer (.pkt) files, device configurations, Wireshark captures, command outputs and screenshots.
- Include a README.md with your name, register number, the operating system, Python version and Packet Tracer (or NS-3) version used, the steps to run each program or open each simulation, and a short note of the result of each task.
- Commit your work progressively – at least one commit for each task. The commit history will be checked; a single upload of all files on the last day will not be accepted.

- Submit the GitHub link below and on the cover page of your report. **An assignment submitted without a working GitHub link will not be evaluated.**

GitHub Repository Link:
https://github.com/niroshinivenkatesan10/CSA0707/tree/main/ASSIGNMENTS

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Scenario — Network for a Smart Manufacturing Plant

An automobile-parts company is building a smart plant. Robots on the shop floor must receive control commands within 10 milliseconds; hundreds of sensors report temperature and vibration; CCTV cameras stream video; office staff use the ERP system and e-mail; and the plant exchanges production data with the head office 400 km away.

The contractor proposes ordinary office Wi-Fi throughout the plant and a single broadband connection to the head office. The plant engineers are worried that electrical noise from welding machines will corrupt data, that robot commands will be delayed, and that the head-office link will be too slow.

The network may grow:

1 production line (40 devices) → 1 plant (1,500 devices) → 5 plants connected to the head office

Each task below gives you data from the plant. Use it to calculate, compare and decide – a correct answer must be supported by your working, not only stated.

Design Challenge:

The company plans to use a private 5G network on the shop floor and to connect autonomous guided vehicles (AGVs).

Re-evaluate your original design and determine whether it should be retained, modified or redesigned.

TASK 1 – Size the Traffic Before Choosing Anything

Bloom's Level: K5 – Evaluate

One production line generates the following traffic:

Source	Quantity	Traffic
Robot controllers	40	200-byte command every 2 ms each
Vibration sensors	600	50 bytes per second each
CCTV cameras	24	4 Mbps each (continuous)
Office users	150	2 Mbps each at peak, 10% active at a time

Work out:

- The total traffic of one line in Mbps, and of the plant if there are 8 lines
- Which source sends the most data and which has the strictest delay requirement – are they the same?
- Whether a single 1 Gbps uplink from the shop floor is enough, and what happens to robot commands if the CCTV traffic shares the same queue
- A ranking of the traffic types by how harmful a 100 ms delay would be, with a reason for each

Alternative Concept Analysis

Suppose the CCTV cameras recorded locally and sent only motion-triggered clips (about 5% of the time) instead of streaming continuously. Recalculate the line traffic, and decide whether the 1 Gbps uplink is still the bottleneck.

Evidence Expected

Complete calculations with units, a delay-versus-volume comparison and a justified ranking.

SOLUTION:

1. Given Data

Source	Quantity	Traffic
Robot controllers	40	200 bytes every 2 ms
Vibration sensors	600	50 bytes/second
CCTV cameras	24	4 Mbps each
Office users	150	2 Mbps each (10% active)

2. Calculate the traffic of one production line

A. Robot Controllers

Traffic per controller:

= (200 × 8) / 0.002

= 800,000 bps

= 0.8 Mbps

For 40 controllers:

= 40 × 0.8

= 32 Mbps

B. Vibration Sensors

= 600 × 50 × 8

= 240,000 bps

= 0.24 Mbps

C. CCTV Cameras

= 24 × 4

= 96 Mbps

D. Office Users

Active users = 150 × 10/100 = 15

Traffic = 15 × 2

= 30 Mbps

3. Total Traffic

Total = 32 + 0.24 + 96 + 30

Total = 158.24 Mbps

For 8 production lines:

= 158.24 × 8

= 1265.92 Mbps

4. Answers

- Highest traffic: CCTV cameras (96 Mbps).
- Strictest delay requirement: Robot controllers (10 ms).
- A 1 Gbps uplink is sufficient for one line.
- A single 1 Gbps uplink is insufficient for eight lines because traffic is 1265.92 Mbps.
- CCTV traffic should not share the same priority queue as robot commands.

5. Ranking by Harmfulness of 100 ms Delay

1. Robot controllers – Can affect machine operation and safety.
2. AGV control – Can delay moving equipment.
3. CCTV cameras – Can cause delayed monitoring.
4. Vibration sensors – Can delay fault detection.
5. Office users – Can cause slow ERP and email access.

6. Alternative Concept Analysis

If CCTV records locally and sends only motion-triggered clips 5% of the time:

$$\text{CCTV traffic} = 96 \times 5/100$$

$$\text{CCTV traffic} = 4.8 \text{ Mbps}$$

New total:

$$= 32 + 0.24 + 4.8 + 30$$

$$= 67.04 \text{ Mbps}$$

For 8 lines:

$$= 67.04 \times 8$$

$$= 536.32 \text{ Mbps}$$

Conclusion:

Local CCTV recording reduces network traffic significantly. The 1 Gbps uplink becomes sufficient for eight lines, assuming no additional traffic.

TASK 2 – Count the Domains in the Contractor's Drawing

Bloom's Level: K5 – Evaluate

The contractor's drawing for the office block shows: three hubs with 8 PCs each, all three hubs connected to one 24-port switch, the switch connected to router interface G0/0, and router interface G0/1 connected to a second switch with 12 IP phones.

Work out:

- The number of collision domains and broadcast domains, with a marked drawing
- The OSI layer at which each device works, and why a hub cannot stop collisions
- Which connections are point-to-point and which are multipoint
- Which standard and which body defines each of: Ethernet (IEEE 802.3), Wi-Fi (IEEE 802.11), IPv4 (IETF RFC 791), the OSI model (ISO 7498), and 5G (3GPP/ITU) – and why the plant should buy only standard-compliant equipment

Alternative Concept Analysis

A colleague proposes replacing the three hubs with three cheap unmanaged switches. Recount the collision and broadcast domains, and state what this change fixes and what it does not.

Evidence Expected

Correct domain counts on a marked drawing, device-layer reasoning and a completed standards table.

SOLUTION :

1. Collision Domains

Given:

- 3 hubs.
- Each hub connects 8 PCs.
- All hubs connect to one switch.
- Switch connects to router G0/0.
- Router G0/1 connects to another switch.
- Second switch connects to 12 IP phones.

Each hub creates one shared collision domain.

First switch:

- 3 hub collision domains.
- 1 router connection.

Second switch:

- 12 IP phone connections.
- 1 router connection.

Total:

3 + 1 + 12 + 1 = 17 collision domains

2. Broadcast Domains

The router separates two broadcast domains.

Therefore:

Total broadcast domains = 2

3. OSI Layers

Device	**OSI Layer**
Hub	Layer 1 – Physical
Switch	Layer 2 – Data Link
Router	Layer 3 – Network
IP Phone	Application and lower layers

A hub cannot stop collisions because it broadcasts incoming signals to all connected devices.

4. Types of Connections

- Point-to-point: Router-to-switch connections.
- Multipoint: Multiple PCs connected through a hub.

5. Standards Table

Technology	Standard	Organization
Ethernet	IEEE 802.3	IEEE
Wi-Fi	IEEE 802.11	IEEE
IPv4	RFC 791	IETF
OSI Model	ISO 7498	ISO
5G	3GPP specifications	3GPP / ITU framework

6. Alternative Concept Analysis

If three hubs are replaced with three unmanaged switches:

Each small switch has:

- 8 PC ports = 8 collision domains.
- 1 uplink port = 1 collision domain.

For three switches:

$3 \times 9 = 27$

Add:

- Main switch to router = 1.
- Second switch phone ports = 12.
- Second switch to router = 1.

Total = 27 + 1 + 12 + 1

= 41 collision domains

Broadcast domains remain 2.

Conclusion:

Replacing hubs with switches reduces collisions and improves network performance. However, switches do not eliminate broadcast traffic.

TASK 3 – Read a Captured Frame Layer by Layer

Bloom's Level: K5 – Evaluate

A capture of one robot command shows the following fields:

```
Ethernet II  Dst: 00:1d:9c:c7:b0:41   Src: 3c:52:82:4e:11:07   Type: 0x0800
IPv4         Src: 10.10.1.25  Dst: 10.10.9.4  TTL: 61  Protocol: 6 (TCP)
TCP          Src Port: 51522  Dst Port: 502 (Modbus)  Seq: 1201  Ack: 88  Len: 12
Data         12 bytes (Modbus write-register command)
```

Work out:

- The OSI layer and the protocol data unit (frame, packet, segment, data) of each line
- How many routers the packet crossed if the controller sets TTL to 64
- The total bytes on the wire (include the 4-byte FCS) and the percentage that is overhead; then repeat for a 1,400-byte camera packet – what does this tell you about small commands?
- Which layer is responsible when (a) the MAC address is wrong, (b) the TTL reaches 0, (c) the destination port is closed
- How the TCP/IP model maps onto these seven layers

Complete the Code

Complete TODO 1 below, and run the program with Python 3.

```
"""How data is wrapped by each layer at the sender and unwrapped at the receiver."""

LAYERS = [("Application", "HTTP"), ("Transport", "TCP"), ("Network", "IP"),
          ("Data link", "ETH")]



def send(message):
pdu = message
    for layer, tag in LAYERS:
        # TODO 1: Wrap pdu with this layer's header "[tag|" and, for the data link layer
        #   only, also a trailer "|FCS]"; for other layers close with "]". Print the
        #   layer name and the size of the new PDU
pass  # replace this line with your code
    return pdu



def receive(frame):
pdu = frame
    for layer, tag in reversed(LAYERS):
        head = "[" + tag + "|"
        tail = "|FCS]" if tag == "ETH" else "]"
        assert pdu.startswith(head) and pdu.endswith(tail), f"{layer}: bad header"
pdu = pdu[len(head):-len(tail)]
    return pdu



frame = send("GET /results")
print("on the wire:", frame)
print("delivered  :", receive(frame))
print(f"overhead  : {len(frame) - len('GET /results')} bytes for 12 bytes of data")
```

Expected output:

```
Application  adds HTTP ->  19 bytes
Transport    adds TCP  ->  25 bytes
Network      adds IP   ->  30 bytes
Data link    adds ETH  ->  40 bytes
on the wire: [ETH|[IP|[TCP|[HTTP|GET /results]]]|FCS]
delivered  : GET /results
overhead   : 28 bytes for 12 bytes of data
```

Change the message to 1,400 bytes and compare the overhead percentage with the 12-byte message.

Alternative Concept Analysis

An industrial protocol could carry the robot command directly inside an Ethernet frame (EtherType 0x8892) with no IP or TCP header. Calculate the bytes saved per command and the commands per second this frees on a 100 Mbps link, and identify what the plant loses by giving up IP.

Evidence Expected

A layer-by-layer reading of the capture, correct overhead calculations and a fault-to-layer mapping.

SOLUTION :

1. Given Data

- Source IP: 10.10.1.25
- Destination IP: 10.10.9.4
- TTL: 61
- Initial TTL: 64
- Protocol: TCP
- Destination port: 502
- Data: 12 bytes

2. OSI Layer Identification

Field	OSI Layer	PDU
Ethernet II	Layer 2	Frame
IPv4	Layer 3	Packet
TCP	Layer 4	Segment
Modbus command	Application layer	Data

3. Number of Routers

Initial TTL = 64

Final TTL = 61

Routers crossed:

64 − 61 = **3 routers**

4. Total Bytes on the Wire

Ethernet header = 14 bytes

IPv4 header = 20 bytes

TCP header = 20 bytes

Data = 12 bytes

FCS = 4 bytes

Total:

= 14 + 20 + 20 + 12 + 4

= **70 bytes**

Overhead:

70 − 12 = 58 bytes

Overhead percentage:

= (58 / 70) × 100

= **82.86%**

5. Camera Packet Comparison

Data = 1400 bytes

Total = 14 + 20 + 20 + 1400 + 4

= 1458 bytes

Overhead = 58 bytes

Overhead percentage:

= (58 / 1458) × 100

= **3.98%**

6. Fault Identification

Problem	Responsible Layer
Wrong MAC address	Data Link
TTL reaches zero	Network
Destination port closed	Transport

7. TCP/IP Mapping

OSI Model	TCP/IP Model
Application	Application
Presentation	Application
Session	Application
Transport	Transport
Network	Internet
Data Link	Network Access
Physical	Network Access

8. Complete the Code – Python Program

```
LAYERS = [
    ("Application", "HTTP"),
    ("Transport", "TCP"),
    ("Network", "IP"),
    ("Data link", "ETH")
]
def send(message):
    pdu = message
    for layer, tag in LAYERS:
        if tag == "ETH":
            pdu = "[" + tag + "|" + pdu + "|FCS]"
        else:
            pdu = "[" + tag + "|" + pdu + "]"
        print(layer, "adds", tag, "->", len(pdu), "bytes")
    return pdu
def receive(frame):
    pdu = frame
    for layer, tag in reversed(LAYERS):
```

```
        head = "[" + tag + "|"
        tail = "|FCS]" if tag == "ETH" else "]"
        assert pdu.startswith(head) and pdu.endswith(tail)
        pdu = pdu[len(head):-len(tail)]
    return pdu
frame = send("GET /results")
print("on the wire:", frame)
print("delivered :", receive(frame))
print("overhead :", len(frame) - len("GET /results"),
    "bytes for 12 bytes of data")
```

9. Expected Output vs Actual Output

Output	Assignment Expected	Program Output
HTTP	19 bytes	19 bytes
TCP	25 bytes	25 bytes
IP	30 bytes	30 bytes
ETH	40 bytes	40 bytes
Overhead	28 bytes	28 bytes

10. Alternative Concept Analysis

If the robot command is directly carried inside an Ethernet frame without IP and TCP:

IP header = 20 bytes

TCP header = 20 bytes

Total saved = 40 bytes per command

At 100 Mbps:

= 100,000,000 / (40 × 8)

= 312,500 saved-header equivalents per second

Disadvantage:

The plant loses IP-based routing and other IP-layer services.

Conclusion:

Small robot commands have a higher overhead percentage than large camera packets.

TASK 4 – Choose a Medium for Each Run (There Is a Trap)

Bloom's Level: K5 – Evaluate

The plant has four cable runs: (a) a robot cell next to arc-welding machines, 85 m; (b) the office floor to the shop-floor switch, 140 m; (c) the shop floor to the data centre, 550 m; (d) the plant to the head office, 400 km.

Work out:

- A medium for each run (Cat6 UTP, Cat6A S/FTP, multimode fibre, single-mode fibre, leased line or wireless) with the deciding reason
- Which run cannot use copper Ethernet at all, and why (look at the distance limits)
- The Shannon capacity of a 250 MHz cable with an SNR of 40 dB and of the same cable near the welders with an SNR of 15 dB – what does the difference mean for the robot cell?
- Why the speed measured on a link is always below the Shannon limit

Complete the Code

Complete TODO 1 and TODO 2 below, and run the program with Python 3.

```
"""Nyquist and Shannon limits for the transmission media proposed for a campus."""
import math



def nyquist(bandwidth_hz, levels):
    # TODO 1: Return the Nyquist maximum bit rate for a noiseless channel: 2 x bandwidth
    #   x log2(levels)
pass  # replace this line with your code



def shannon(bandwidth_hz, snr_db):
    # TODO 2: Convert the SNR from decibels to a ratio (10 ** (dB / 10)) and return the
    #   Shannon capacity: bandwidth x log2(1 + SNR)
pass  # replace this line with your code



media = [("Telephone line", 3_100, 30), ("Cat6 pair", 250e6, 40),
         ("Wi-Fi 6 channel", 160e6, 25)]
for name, bw, snr in media:
    print(f"{name:<16} Shannon limit {shannon(bw, snr) / 1e6:10.2f} Mbps")
print(f"Nyquist, 3,100 Hz with 4 levels: {nyquist(3_100, 4) / 1e3:.1f} kbps")
```

Expected output:

```
Telephone line   Shannon limit       0.03 Mbps
Cat6 pair        Shannon limit    3321.96 Mbps
Wi-Fi 6 channel  Shannon limit    1329.50 Mbps
Nyquist, 3,100 Hz with 4 levels: 12.4 kbps
```

Find the SNR (in dB) a 3,100 Hz telephone line would need to carry 56 kbps, and judge whether that is realistic.

Alternative Concept Analysis

What if the robot cell used Wi-Fi 6 instead of shielded cable? Using the welding-area SNR of 15 dB on a 160 MHz channel, calculate the Shannon limit, and decide whether a wireless link is acceptable for 10 ms robot commands.

Evidence Expected

A justified medium for every run, the trapped run identified, and Shannon calculations with interpretation.

SOLUTION :

1. Transmission Media Selection

Connection	**Distance**	**Recommended Medium**
Robot cell near welding machines	85 m	Cat6A S/FTP
Office to shop-floor switch	140 m	Fibre optic
Shop floor to data centre	550 m	Multimode fibre
Plant to head office	400 km	Single-mode fibre / leased line

Important:

The 140 m Ethernet copper run exceeds the normal 100 m Ethernet channel limit. Therefore, use fibre or an intermediate switch.

2. Shannon Capacity Formula

$C = B \log_2(1 + SNR)$

Where:

- C = Channel capacity.
- B = Bandwidth.
- SNR = Signal-to-noise ratio.

Convert decibels:

SNR = 10^(dB/10)

3. Calculate at 40 dB

Given:

B = 250 MHz

SNR = 10^4

$C = 250 \times 10^6 \times \log_2(1 + 10000)$

C = 3321.96 Mbps

4. Calculate at 15 dB

SNR = 10^(15/10) => 31.62

$C = 250 \times 10^6 \times \log_2(1 + 31.62)$

C ≈ 1256.95 Mbps

5. Complete the Code

```
import math
def nyquist(bandwidth_hz, levels):
    return 2 * bandwidth_hz * math.log2(levels)
def shannon(bandwidth_hz, snr_db):
    snr = 10 ** (snr_db / 10)
    return bandwidth_hz * math.log2(1 + snr)
media = [
    ("Telephone line", 3100, 30),
    ("Cat6 pair", 250e6, 40),
    ("Wi-Fi 6 channel", 160e6, 25)
]
for name, bw, snr in media:
    print(f"{name:<16} Shannon limit {shannon(bw, snr) / 1e6:10.2f} Mbps")
print(f"Nyquist, 3,100 Hz with 4 levels: {nyquist(3100, 4) / 1e3:.1f} kbps")
```

6. Expected Output vs Actual Output

Output	Expected	Actual
Telephone line	0.03 Mbps	0.03 Mbps
Cat6 pair	3321.96 Mbps	3321.96 Mbps
Wi-Fi 6 channel	1329.50 Mbps	1329.50 Mbps
Nyquist	12.4 kbps	12.4 kbps

7. Additional Calculation

For a telephone line to carry 56 kbps:

$C = B \log_2(1 + SNR)$

SNR = 2^(56000/3100) − 1

Required SNR ≈ 2.74×10^5

SNR in dB ≈ 54.38 dB

This is very high for an ordinary telephone line.

8. Alternative Concept Analysis

Wi-Fi 6 at 15 dB SNR:

$C = 160 \times 10^6 \times \log_2(1 + 31.62)$

C = 804.45 Mbps

Although the theoretical capacity is high, Wi-Fi cannot guarantee the 10 ms robot-command deadline because of interference, contention and variable latency.

Conclusion:

Lower SNR reduces the theoretical channel capacity. Welding interference makes shielded cable or fibre preferable for critical industrial communication.

TASK 5 – Wi-Fi 6 or 5G for Moving Equipment?

Bloom's Level: K5 – Evaluate

An AGV travels at 2 m/s along a 1.2 km route. Wi-Fi access points are 30 m apart, and a Wi-Fi hand-over takes about 150 ms. A private 5G cell covers 300 m, and a 5G hand-over takes about 20 ms. AGV commands have a 10 ms deadline.

Work out:

- The number of hand-overs per trip for Wi-Fi and for 5G, and the total time per trip during which the AGV is not reachable
- Which of the three 5G service types (eMBB, URLLC, mMTC) the AGVs, the cameras and the sensors each need
- Where Bluetooth, Zigbee or LoRa would be a better choice than Wi-Fi or 5G in the plant
- Whether Wi-Fi 6 with fast roaming (802.11r) could meet the deadline – and what remains risky

Alternative Concept Analysis

The sensors could use LoRaWAN (a few kbps, kilometres of range, years on a battery) instead of Wi-Fi. Calculate whether 600 sensors at 50 bytes per second fit within one LoRa gateway's capacity of roughly 5 kbps, and decide which sensors should stay on Wi-Fi.

Evidence Expected

Hand-over calculations, a mapping of traffic to 5G service types and a justified wireless decision.

SOLUTION :

1. Given Data

- AGV speed = 2 m/s
- Route = 1.2 km
- Wi-Fi AP distance = 30 m
- Wi-Fi handover = 150 ms
- 5G cell coverage = 300 m
- 5G handover = 20 ms

- Command deadline = 10 ms

2. Wi-Fi Handover Calculation

Number of coverage sections:

$$1200 / 30 = 40$$

Handover count = 40 − 1

Handover count = 39 handovers

Total unavailable time:

$$= 39 \times 150 \text{ ms}$$

$$= 5850 \text{ ms}$$

$$= 5.85 \text{ seconds}$$

3. 5G Handover Calculation

Number of coverage sections:

$$1200 / 300 = 4$$

Handover count = 4 − 1

Handover count = 3 handovers

Total unavailable time:

$$= 3 \times 20 \text{ ms}$$

$$= 60 \text{ ms}$$

4. Comparison

Parameter	**Wi-Fi 6**	**Private 5G**
Handover count	39	3
Handover time	150 ms	20 ms
Total unavailable time	5850 ms	60 ms

10 ms deadline	Not suitable with given handover	Better, but not guaranteed

5. 5G Service Types

Service	Application
eMBB	CCTV cameras
URLLC	Robots and AGVs
mMTC	Large numbers of sensors

6. Wireless Technology Selection

- Bluetooth: Short-range equipment communication.
- Zigbee: Low-power sensor networks.
- LoRa: Long-range, low-data-rate sensors.
- Wi-Fi 6: Office users and general wireless access.
- 5G: Mobile robots and industrial control where suitable.

7. Alternative Concept Analysis

LoRaWAN capacity:

= 600 × 50 bytes × 8

= 240,000 bps

= 240 kbps

Available capacity = 5 kbps.

240 / 5 = 48

Therefore, the sensors require approximately 48 times the available gateway capacity.

Conclusion:

All 600 sensors cannot transmit their stated traffic through one 5 kbps LoRa gateway. Low-rate sensors can use LoRa, while high-rate sensors should remain on Wi-Fi or wired connections.

TASK 6 – Interpret an Ethernet Port Report

Bloom's Level: K5 – Evaluate

A shop-floor PC is slow. Its network interface reports:

```
Speed: 100Mb/s   Duplex: Half   Auto-negotiation: off
Link detected: yes   rx_errors: 2381   late_collisions: 412   crc_errors: 1907
```

Work out:

- What a late collision and a CRC error each mean, and what the combination suggests about the two ends of the link
- The likely cause (a duplex mismatch with a gigabit switch port set to auto) and the exact configuration change on Linux (ethtool) or Windows
- Whether the MAC address 01:00:5e:0a:01:02 belongs to a single device or a group, and what the first three bytes of 3c:52:82:4e:11:07 identify
- Whether jumbo frames (9,000 bytes) would help the CCTV storage server, and what must also change for them to work

Alternative Concept Analysis

Compare a shop floor wired as a ring of industrial switches with fast ring recovery against a star of switches. For one cable cut, state the recovery time each needs and which devices lose connection.

Evidence Expected

A reasoned interpretation of the report, the corrective configuration and correct answers on addressing and frame size.

SOLUTION :

1. Given Data

Parameter	**Value**
Speed	100 Mbps
Duplex	Half
Auto-negotiation	Off
RX Errors	2381
Late Collisions	412
CRC Errors	1907

2. Error Explanation

Late Collision: A collision detected after the normal collision-detection window. It may indicate duplex mismatch or cabling problems.

CRC Error: A frame fails its cyclic redundancy check, indicating corrupted data.

3. Likely Cause

The likely problem is a duplex mismatch between the PC and switch.

One side uses half-duplex while the other side may use full-duplex.

4. Linux Configuration

Check interface:

```
ethtool eth0
```

Configure:

```
sudo ethtool -s eth0 speed 100 duplex full autoneg off
```

The switch port must also be configured for 100 Mbps full-duplex.

5. MAC Address Analysis

MAC address:

01:00:5e:0a:01:02

This is an IPv4 multicast MAC address.

It represents a group, not one individual device.

MAC address:

3c:52:82:4e:11:07

The first three bytes are:

3C:52:82

They identify the manufacturer through its assigned OUI.

6. Jumbo Frames

Jumbo frames support approximately 9000-byte payloads.

Advantages:

- Reduce packet overhead.
- Improve large data transfers.
- Help CCTV storage servers.

Requirement:

All devices in the communication path must support and be configured for compatible jumbo frame sizes.

7. Alternative Concept Analysis

Ring Topology	**Star Topology**
Provides an alternative path	Uses a central switch
Can recover from a cable cut	A cut can isolate connected devices

Recovery depends on ring protocol	Recovery depends on redundancy and switch configuration

Fast industrial ring recovery may take milliseconds, depending on the protocol and equipment. A simple star without redundant links cannot automatically recover from a failed link.

TASK 7 – Will the Receiver Notice the Error?

Bloom's Level: K5 – Evaluate

Frames between two shop-floor switches are protected by a CRC with generator 1011 ($x^3 + x + 1$).

Work out:

- The CRC bits for the data 1101011011, by hand (show the division)
- Whether a received frame 1101011011101 is accepted or rejected
- The longest burst error this generator is certain to detect, and an error pattern it cannot detect
- The two-dimensional parity of a 4 × 4 block of your choice, and a 4-bit error pattern that two-dimensional parity misses
- The Internet checksum of the four 16-bit words 4500, 0073, 0000, 4000 (hexadecimal)

Complete the Code

Complete TODO 1 below to finish the modulo-2 division, and run the program with Python 3.

```
"""CRC error detection for frames sent between two switches."""


def mod2_div(dividend, divisor):
    """Return the remainder of modulo-2 (XOR) division of two bit strings."""
    rem = list(dividend)
    for i in range(len(dividend) - len(divisor) + 1):
        # TODO 1: If the current leading bit rem[i] is 1, XOR the divisor into rem
        #   starting at position i
pass  # replace this line with your code
    return "".join(rem[-(len(divisor) - 1):])


def encode(data, gen):
crc = mod2_div(data + "0" * (len(gen) - 1), gen)
    return data + crc


def check(frame, gen):
    return set(mod2_div(frame, gen)) == {"0"}


GEN = "1011"                            # x^3 + x + 1
frame = encode("1101011011", GEN)
print("transmitted:", frame)
print("no error   :", check(frame, GEN))
bad = frame[:4] + ("1" if frame[4] == "0" else "0") + frame[5:]
print("1-bit error:", check(bad, GEN))
burst = frame[:2] + "".join("1" if b == "0" else "0" for b in frame[2:5]) + frame[5:]
print("3-bit burst:", check(burst, GEN))
```

Expected output:

```
transmitted: 1101011011100
no error   : True
1-bit error: False
3-bit burst: False
```

Find, with the program, an error pattern that generator 1011 cannot detect, and explain why.

Alternative Concept Analysis

Which errors would a simple even-parity bit catch that the CRC-3 generator 1011 misses, and which would it miss? Give one 2-bit and one 4-bit error pattern for each method and state the result.

Evidence Expected

Hand calculations with working, an undetected-error example for each method and a comparison of their strength.

SOLUTION :

1. Given Data

Data = 1101011011

Generator = 1011

Generator polynomial = $x^3 + x + 1$

2. CRC Calculation

Append three zeros:

1101011011 000

Perform modulo-2 division using XOR.

Final CRC remainder:

100

Transmitted frame:

1101011011100

3. Received Frame

Received:

1101011011101

The remainder is non-zero.

Therefore:

Frame is rejected.

4. Longest Burst Error

Generator length = 4 bits.

Maximum guaranteed detectable burst length:

4 − 1 = 3 bits

An undetectable error example is:

1011000000000

This is a multiple of the generator polynomial, so its CRC remainder is zero.

5. Complete the Code

```
def mod2_div(dividend, divisor):
    rem = list(dividend)
    for i in range(len(dividend) - len(divisor) + 1):
        if rem[i] == "1":
            for j in range(len(divisor)):
                rem[i + j] = str(
                    int(rem[i + j]) ^ int(divisor[j])
                )
    return "".join(rem[-(len(divisor) - 1):])
def encode(data, gen):
    crc = mod2_div(data + "0" * (len(gen) - 1), gen)
```

```
    return data + crc

def check(frame, gen):
    return set(mod2_div(frame, gen)) == {"0"}

GEN = "1011"
frame = encode("1101011011", GEN)
print("transmitted:", frame)
print("no error :", check(frame, GEN))
bad = frame[:4] + ("1" if frame[4] == "0" else "0") + frame[5:]
print("1-bit error:", check(bad, GEN))
burst = frame[:2] + "".join(
    "1" if b == "0" else "0" for b in frame[2:5]
) + frame[5:]
print("3-bit burst:", check(burst, GEN))
```

6. Expected Output vs Actual Output

Output	**Expected**	**Actual**
Transmitted	1101011011100	1101011011100
No error	True	True
1-bit error	False	False
3-bit burst	False	False

7. Two-Dimensional Parity

Example data:

1 0 1 1

0 1 0 1

1 1 0 0

0 0 1 1

Row parity:

1

0

0

0

Column parity:

0 0 0 1

A four-bit rectangular error pattern can escape two-dimensional parity because every affected row and column contains an even number of flipped bits.

8. Internet Checksum

Given:

4500, 0073, 0000, 4000

Hexadecimal sum:

4500 + 0073 + 0000 + 4000

= 8573

One's complement:

FFFF − 8573

= 7A8C

9. Alternative Concept Analysis

Error Pattern	**CRC-3**	**Even Parity**
1000	Detected	Detected
1100	Detected	Not detected
1011000000000	Not detected	Depends on parity of the full error pattern

CRC is generally stronger for detecting burst errors.

TASK 8 – Correct the Error or Ask Again?

Bloom's Level: K5 – Evaluate

A vibration sensor sends 16-bit readings over a noisy radio link where retransmission takes 40 ms.

Work out:

- The number of Hamming check bits needed for a 16-bit word (use $2^r \geq m + r + 1$) and the overhead percentage
- The minimum Hamming distance of the code {00000, 01011, 10101, 11110}, and how many errors it can detect and correct
- The corrected data when the Hamming(7,4) codeword 0110011 arrives as 0110111
- Whether forward error correction or retransmission (ARQ) suits this link, and why video streaming and 5G use FEC

Complete the Code

Complete TODO 1 below, and run the program with Python 3.

```
"""Hamming(7,4) code: correct a single-bit error in a 4-bit sensor reading."""



def encode(d):                  # d = [d1, d2, d3, d4]
    p1 = d[0] ^ d[1] ^ d[3]
    p2 = d[0] ^ d[2] ^ d[3]
    p3 = d[1] ^ d[2] ^ d[3]
    return [p1, p2, d[0], p3, d[1], d[2], d[3]]      # positions 1..7



def decode(c):
    # TODO 1: Compute the syndrome bits s1 (positions 1,3,5,7), s2 (2,3,6,7) and s3
    #   (4,5,6,7); pos = s1 + 2*s2 + 4*s3. If pos is not 0, flip bit pos (index pos-1).
    #   Return pos and the data bits at positions 3, 5, 6, 7
pass  # replace this line with your code



word = encode([1, 0, 1, 1])
print("codeword       :", "".join(map(str, word)))
for flip in (None, 5):
rx = word[:]
    if flip:
rx[flip - 1] ^= 1
pos, data = decode(rx)
print(f"error at {str(flip):<4}: syndrome {pos}, data {''.join(map(str, data))}")
```

Expected output:

```
codeword       : 0110011
error at None: syndrome 0, data 1011
error at 5   : syndrome 5, data 1011
```

Flip two bits of the codeword and show that the decoder now “corrects” the wrong bit.

Alternative Concept Analysis

At a bit-error rate of 1 in 1,000, compare sending 16-bit readings with Hamming correction against retransmitting on error: calculate the bits sent per reading and the average delay for each, using the 40 ms retransmission time.

Evidence Expected

Correct calculations, a corrected codeword and a justified choice between FEC and ARQ.

SOLUTION :

1. Given Data

Data bits = 16

Formula:

$2^r \geq m + r + 1$

2. Calculate Hamming Check Bits

For r = 4:

$2^4 = 16$

$16 < 16 + 4 + 1$

$16 < 21$ (not sufficient)

For r = 5:

$2^5 = 32$

$32 \geq 16 + 5 + 1$

Therefore:

Required check bits = 5

Total bits = 16 + 5 = 21

Overhead percentage:

$= (5 / 21) \times 100$

$= 23.81\%$

3. Minimum Hamming Distance

Given:

00000, 01011, 10101, 11110

Pairwise Hamming distances:

- 00000 and 01011 = 3
- 00000 and 10101 = 3
- 00000 and 11110 = 4
- 01011 and 10101 = 4
- 01011 and 11110 = 3
- 10101 and 11110 = 3

Minimum distance = **3**

Error detection = 2 bits.

Error correction = 1 bit.

4. Hamming Code Correction

Original codeword:

0110011

Received codeword:

0110111

The error is at position 5.

Syndrome = 5

Flip bit 5 to correct the codeword.

Corrected codeword:

0110011

Extracted data:

1011

5. Complete the Code

```
def encode(d):
    p1 = d[0] ^ d[1] ^ d[3]
    p2 = d[0] ^ d[2] ^ d[3]
    p3 = d[1] ^ d[2] ^ d[3]
    return [p1, p2, d[0], p3, d[1], d[2], d[3]]
def decode(c):
    s1 = c[0] ^ c[2] ^ c[4] ^ c[6]
    s2 = c[1] ^ c[2] ^ c[5] ^ c[6]
    s3 = c[3] ^ c[4] ^ c[5] ^ c[6]
    pos = s1 + 2 * s2 + 4 * s3
    if pos != 0:
        c[pos - 1] ^= 1
    data = [c[2], c[4], c[5], c[6]]
    return pos, data
word = encode([1, 0, 1, 1])
print("codeword :", "".join(map(str, word)))
for flip in (None, 5):
    rx = word[:]
    if flip:
        rx[flip - 1] ^= 1
    pos, data = decode(rx)
    print(f"error at {str(flip):<4}: syndrome {pos}, data {''.join(map(str, data))}")
```

6. Expected Output vs Actual Output

Output	Expected	Actual
Codeword	0110011	0110011
No error syndrome	0	0
No error data	1011	1011
Error position	5	5
Corrected data	1011	1011

7. Alternative Concept Analysis

Bit error rate = 1/1000 = 0.001.

Probability of an error in a 16-bit reading:

$$P = 1 - (0.999)^{16}$$

$$\approx 0.01588$$

$$= 1.588\%$$

Hamming transmission:

16 + 5 = 21 bits per reading.

Expected retransmission bits without FEC:

$$16 / (1 - 0.01588)$$

$$\approx 16.26 \text{ bits per successful reading.}$$

FEC is preferable when retransmission takes 40 ms and low latency is important.

Conclusion:

Forward Error Correction is more suitable for the noisy sensor link because it corrects single-bit errors without waiting for retransmission.

TASK 9 – How Big Must the Window Be?

Bloom's Level: K5 – Evaluate

The head-office link is 400 km of fibre at 1 Gbps; frames are 1,500 bytes; light travels at 2×10^8 m/s in fibre.

Work out:

- The transmission time, the propagation time and the ratio a = propagation / transmission
- The link utilisation with stop-and-wait ARQ – and what it means in Mbps
- The smallest window that keeps the link fully used, and the number of sequence-number bits needed for Go-Back-N and for Selective Repeat with that window
- Which of the two ARQ protocols resends fewer frames after one loss, and what it costs the receiver
- How HDLC or PPP frames mark the start and end of a frame

Complete the Code

Complete TODO 1 below, and run the program with Python 3.

```
"""Link utilisation of stop-and-wait and sliding-window protocols on different links."""


def utilisation(frame_bits, rate_bps, distance_m, window, speed=2e8):
tx = frame_bits / rate_bps                # transmission time
    prop = distance_m / speed                 # propagation time
    # TODO 1: a = prop / tx. Utilisation is window / (1 + 2a), but never more than 1.
    #   Return the utilisation and a
pass  # replace this line with your code



links = [("Campus LAN, 200 m, 1 Gbps", 1e9, 200),
         ("City fibre, 50 km, 1 Gbps", 1e9, 50_000),
         ("Satellite, 36,000 km, 10 Mbps", 10e6, 36_000_000)]
for name, rate, dist in links:
    u1, a = utilisation(12_000, rate, dist, 1)
    u7, _ = utilisation(12_000, rate, dist, 7)
    u127, _ = utilisation(12_000, rate, dist, 127)
    print(f"{name:<31} a = {a:7.2f}  stop-and-wait {u1:6.1%}  "
f"W=7 {u7:6.1%}  W=127 {u127:6.1%}")
```

Expected output:

```
Campus LAN, 200 m, 1 Gbps       a =    0.08  stop-and-wait  85.7%  W=7 100.0%  W=127 100.0%
City fibre, 50 km, 1 Gbps       a =   20.83  stop-and-wait   2.3%  W=7  16.4%  W=127 100.0%
Satellite, 36,000 km, 10 Mbps   a =  150.00  stop-and-wait   0.3%  W=7   2.3%  W=127 42.2%
```

Calculate the smallest window that gives 100% utilisation on the satellite link.

Alternative Concept Analysis

Suppose the head-office link used Selective Repeat with a window of 64 instead of the window you calculated. Calculate the resulting utilisation and throughput, and the receiver buffer space needed.

Evidence Expected

Correct timing, utilisation and window calculations, sequence-number sizes and a justified protocol choice.

SOLUTION :

1. Given Data

- Distance = 400 km = 400,000 m
- Rate = 1 Gbps
- Frame size = 1500 bytes
- Propagation speed = 2×10^8 m/s

2. Transmission Time

Frame bits:

$1500 \times 8 = 12{,}000$ bits

Transmission time:

$= 12{,}000 / 1{,}000{,}000{,}000$

$= 12$ microseconds

3. Propagation Time

$= 400{,}000 / (2 \times 10^8)$

$= 0.002$ seconds

$= 2$ ms

4. Calculate a

a = Propagation time / Transmission time

a = 0.002 / 0.000012

a = 166.67

5. Stop-and-Wait Utilization

U = 1 / (1 + 2a)

U = 1 / (1 + 333.33)

U = 0.002991

U = **0.299%**

Throughput:

$= 1 \text{ Gbps} \times 0.002991$

$= 2.99 \text{ Mbps}$

6. Minimum Window Size

$W = 1 + 2a$

$W = 334.33$

Round up:

$W = 335 \text{ frames}$

7. Sequence Number Bits

Go-Back-N

Maximum window = $2^n - 1$

For 335 frames:

$2^9 - 1 = 511$

Therefore, 9 bits.

Selective Repeat

Maximum window = 2^{n-1}

For 335 frames:

$2^{10-1} = 512$

Therefore, 10 bits.

8. Complete the Code

```
def utilisation(frame_bits, rate_bps, distance_m, window, speed=2e8):
    tx = frame_bits / rate_bps
    prop = distance_m / speed
    a = prop / tx
    u = min(1, window / (1 + 2 * a))
    return u, a
links = [
    ("Campus LAN, 200 m, 1 Gbps", 1e9, 200),
    ("City fibre, 50 km, 1 Gbps", 1e9, 50000),
    ("Satellite, 36,000 km, 10 Mbps", 10e6, 36000000)
]
for name, rate, dist in links:
    u1, a = utilisation(12000, rate, dist, 1)
    u7, _ = utilisation(12000, rate, dist, 7)
```

```
    u127, _ = utilisation(12000, rate, dist, 127)
    print(f"{name:<31} a = {a:7.2f} stop-and-wait {u1:6.1%} W=7 {u7:6.1%} W=127
{u127:6.1%}")
```

9. Expected Output vs Actual Output

Link	Expected Utilization	Actual Utilization
Campus Stop-and-Wait	85.7%	85.7%
Campus W=7	100.0%	100.0%
Campus W=127	100.0%	100.0%
City Stop-and-Wait	2.3%	2.3%
City W=7	16.4%	16.4%
City W=127	100.0%	100.0%
Satellite Stop-and-Wait	0.3%	0.3%
Satellite W=7	2.3%	2.3%
Satellite W=127	42.2%	42.2%

10. Satellite Minimum Window

a = 150

W = 1 + 2 × 150

W = 301 frames

11. Alternative Concept Analysis

Selective Repeat window = 64

Utilization:

64 / 334.33

= 19.14%

Throughput:

1 Gbps × 0.1914

= 191.43 Mbps

Receiver buffer:

64 × 1500

= 96,000 bytes

= 96 KB

Selective Repeat retransmits only lost frames, while Go-Back-N may retransmit multiple subsequent frames.

TASK 10 – Find Where the Delay Comes From

Bloom's Level: K5 – Evaluate

A traceroute from the plant to the head-office server gives:

```
1  10.10.1.1       0.6 ms   0.5 ms   0.6 ms
2  100.64.0.1      3.1 ms   2.9 ms   3.4 ms
3  172.31.4.9      4.0 ms  38.2ms    4.2 ms
4  103.21.8.17     11.8 ms  11.5ms   12.1ms
5  * * *
6  10.200.0.10     12.6 ms  12.4ms   12.9ms
```

Work out:

- The propagation delay you would expect for 400 km of fibre, and how much of the measured delay it explains
- Why hop 3 shows one value of 38.2 ms, and why hop 5 shows only stars – are they faults?
- Which hop adds the most delay
- Your own traceroute, ping and Wireshark capture of a real connection, with the headers of each layer identified

Alternative Concept Analysis

Compare continuous monitoring with SNMP counters and flow records (NetFlow) against occasional ping and traceroute tests: which of the problems in the traceroute above would each have detected, and how quickly?

Evidence Expected

A reasoned reading of the traceroute, a propagation-delay check and your own measured evidence.

SOLUTION :

1. Given Traceroute

Hop	IP Address	Delay
1	10.10.1.1	0.6 ms
2	100.64.0.1	3.1 ms
3	172.31.4.9	4.0 ms, 38.2 ms, 4.2 ms
4	103.21.8.17	11.8 ms
5	* * *	No response
6	10.200.0.10	12.6 ms

2. Expected Propagation Delay

Distance = 400 km

Propagation speed = 2×10^8 m/s

Delay:

$= 400{,}000 / (2 \times 10^8)$

$= 0.002$ seconds

$= 2$ ms one-way

Approximately 4 ms round-trip.

3. Why Hop 3 Shows 38.2 ms?

Possible reasons:

- Router processing delay.
- Temporary congestion.
- ICMP response rate limiting.

It does not necessarily indicate a network fault.

4. Why Hop 5 Shows Stars?

Possible reasons:

- Firewall blocks ICMP.
- Router does not respond to traceroute.
- ICMP rate limiting.

Stars do not always indicate a failed router.

5. Which Hop Adds the Most Delay?

Using representative RTT values:

Hop 1 = 0.6 ms

Hop 2 additional delay = 3.1 − 0.6 = 2.5 ms

Hop 3 additional delay = 4.0 − 3.1 = 0.9 ms

Hop 4 additional delay = 11.8 − 4.0 = 7.8 ms

Hop 6 additional delay = 12.6 − 11.8 = 0.8 ms

Therefore:

Hop 4 adds the most delay (approximately 7.8 ms).

6. Commands for Practical Evidence

Windows:

```
tracert google.com

ping google.com
```

Linux:

```
traceroute google.com

ping google.com
```

Wireshark can be used to capture and inspect:

- Ethernet headers.
- IPv4 headers.

- TCP headers.
- Application data.

7. Alternative Concept Analysis

SNMP and NetFlow	**Ping and Traceroute**
Continuous monitoring	Occasional testing
Detects traffic changes	Measures connectivity
Detects interface errors	Helps identify route problems
Provides traffic statistics	Shows individual hop delays

Conclusion:

Continuous monitoring provides better visibility into network performance than occasional testing alone.

TASK 11 – Defend the Network Design

Bloom's Level: K5 – Evaluate

Recommend the network design for the plant.

Work out:

- The medium, wireless technology and Ethernet settings for each area
- The error-control and flow-control method for each link, with the calculation that justifies it
- The two biggest risks that remain in your design
- Whether your design still works for five plants

Your recommendation must quote **numbers from Tasks 1–10**.

Alternative Concept Analysis

Suppose the whole plant used private 5G instead of a wired backbone with Wi-Fi. Using your traffic figures from Task 1, decide which traffic 5G should carry and which must stay on cable, and justify the split.

Evidence Expected

A recommendation supported by your own calculations and measurements.

SOLUTION :

1. Recommended Network Design

Area	**Recommended Technology**
Robot controllers	Shielded Ethernet / private 5G URLLC where validated
Vibration sensors	Industrial Ethernet / suitable wireless sensor network
CCTV cameras	Wired Ethernet
Office users	Wi-Fi 6
Plant backbone	Fibre optic
Head office	Single-mode fibre / leased line

2. Error Control and Flow Control

Network	Recommended Method
Ethernet	CRC
Wireless sensors	FEC or ARQ depending on delay
TCP connections	Sliding Window
Robot commands	Industrial real-time communication with redundancy
CCTV	TCP or suitable streaming transport

3. Two Biggest Risks

1. Wireless interference from welding machines can affect communication reliability.
2. High CCTV traffic can create congestion and delay critical robot commands.

4. Five-Plant Network

For five plants:

Each plant traffic = 1265.92 Mbps.

Total:

$= 1265.92 \times 5$

$= 6329.6$ Mbps

$= 6.33$ Gbps

Therefore, a single 1 Gbps head-office connection is insufficient.

A high-capacity fibre backbone with suitable redundancy and traffic management is recommended.

5. Alternative Concept Analysis

If the entire plant uses private 5G:

Suitable:

- AGVs.
- Mobile robots.

- Wireless sensors.
- Mobile industrial equipment.

Better kept wired:

- CCTV storage servers.
- Fixed high-bandwidth systems.
- Core switches.
- Data centre connections.

Conclusion:

A hybrid network combining wired Ethernet, fibre, Wi-Fi 6 and private 5G provides better performance and reliability than using only one technology.

TASK 12 – Re-evaluate and Adapt the Design

Bloom's Level: K5 – Evaluate

Changed Requirement

The company adds a **private 5G network** on the shop floor and **50 AGVs** that move parts between lines and must never miss a command.

Re-evaluate your original design. You may **RETAIN → MODIFY → or REDESIGN** your original solution.

A. Impact Analysis

Identify which media, wireless choices and error-control settings are affected, and recalculate the traffic from Task 1 with the AGVs added.

Evidence Expected

A quantified analysis of how the changes affect the original design.

B. Test the Original Design

Show, using your hand-over and delay calculations, whether the original design can serve 50 moving AGVs.

Evidence Expected

A calculation-based test of the original design.

C. Adaptation or Redesign Describe the revised design (for example: 5G URLLC for AGVs and robots, Wi-Fi 6 for offices, fibre backbone).

Evidence Expected

An appropriate design response.

D. Evidence-Based Defence

Defend your decision with numbers and the trade-offs involved.

Evidence Expected

A well-supported final decision.

SOLUTION :

A. Impact Analysis

The company adds:

- Private 5G network.
- 50 AGVs.
- Mobile part transportation.
- 10 ms command deadline.

Original traffic:

158.24 Mbps per line.

For eight lines:

1265.92 Mbps.

B. Test the Original Design

Wi-Fi handover:

1200 / 30 = 40 coverage sections.

Handovers = 39.

Total interruption:

= 39 × 150

= 5850 ms

5G handover:

1200 / 300 = 4 coverage sections.

Handovers = 3.

Total interruption:

= 3 × 20

= 60 ms

Comparison

Parameter	Wi-Fi 6	Private 5G
Handovers	39	3
Interruption	5850 ms	60 ms
Deadline	10 ms	10 ms
Suitability	Poor with given handover time	Better, but requires validation

The original Wi-Fi design cannot reliably serve 50 moving AGVs with the given handover time.

The given 5G handover time is also greater than the 10 ms command deadline. Therefore, additional reliability and mobility testing is required.

C. Adaptation or Redesign

Revised Network Design

Area	Revised Technology
AGVs	Private 5G URLLC
Robot controllers	Industrial Ethernet / URLLC
CCTV	Wired Ethernet
Office systems	Wi-Fi 6
Sensors	Industrial Ethernet / LoRaWAN for suitable low-rate sensors
Backbone	High-speed fibre
Head office	Redundant single-mode fibre

D. Evidence-Based Defence

Reasons for redesign:

1. Original plant traffic is 1265.92 Mbps.
2. Additional assumed AGV traffic is 40 Mbps.
3. Revised traffic is 1305.92 Mbps.

4. Original 1 Gbps uplink cannot handle the revised traffic.
5. Wi-Fi handover interruption is 5850 ms.
6. Private 5G reduces the calculated interruption to 60 ms.
7. Fibre provides high-speed and reliable backbone connectivity.
8. Traffic prioritization protects critical robot commands from CCTV congestion.

Final Recommendation

Decision: MODIFY THE ORIGINAL DESIGN.

The company should adopt a hybrid network with private 5G for AGVs, industrial Ethernet for fixed robot controllers, Wi-Fi 6 for office users and fibre for the backbone.

This design improves mobility, reduces congestion and supports future expansion.